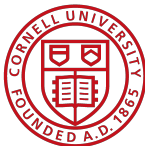


Causal Mediation in Natural Experiments

Senan Hogan-Hennessy
Economics Department, Cornell University
seh325@cornell.edu



Cornell Economics, PhD Defence
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Introduction

Motivation:

- Causal Mediation (CM) is a framework for studying mechanisms of causal effects, but is not widely used in economics
- Economists stick to *suggestive evidence*, leaving mechanisms not identified.

Contribution:

- Develop selection bias concept for CM, crystallising reasoning for economists dismissal of these methods
- Develop a Marginal Treatment Effect (MTE) approach to credibly estimate causal channels (direct and indirect effects).

Key findings:

- Apply methods to the Oregon Health Insurance Experiment, studying mediating role of healthcare take-up
- Bring CM into economic quasi-experimental causal inference.

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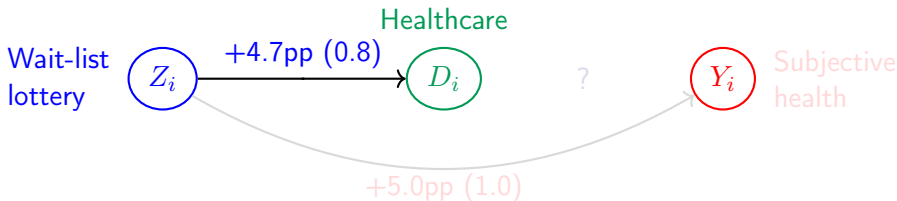
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1. Suggestive Evidence

In 2008, Oregon gave access to socialised health insurance by wait-list lottery (Finkelstein et al, 2012).

Figure: Model for Suggestive Evidence of a Mechanism.



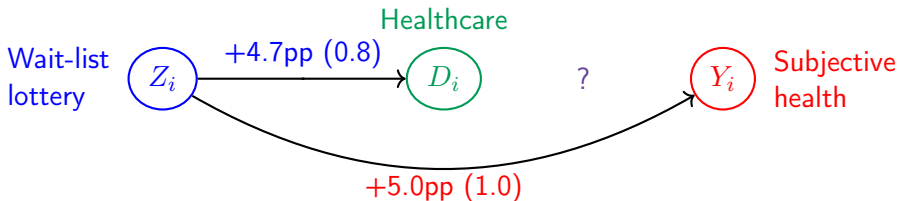
Necessary but not sufficient evidence on the mediating mechanism:

- Is $D_i \rightarrow Y_i$ small, large, non-zero?
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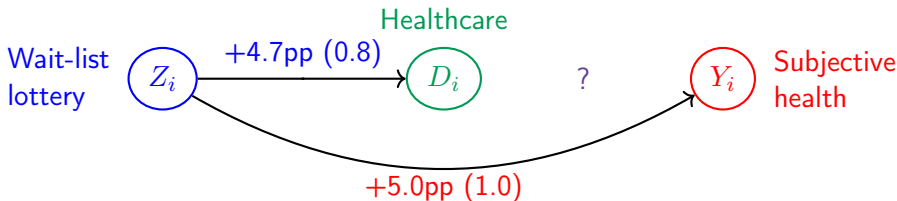
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2. Causal Mediation

Causal Mediation (CM) models the entire system, giving sufficient evidence on the mediating mechanism.



Average Direct Effect (ADE) and Average Indirect Effect (AIE):

- ADE is causal effect $Z_i \rightarrow Y_i$
- AIE is causal effect of $D_i(Z_i) \rightarrow Y_i$.

Identification: Requires mediator D_i is quasi-randomly assigned. . . .

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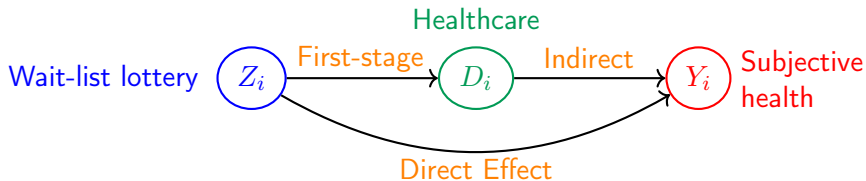
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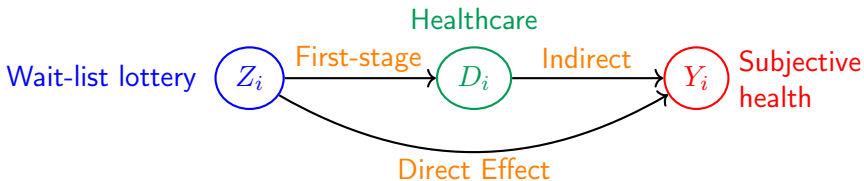
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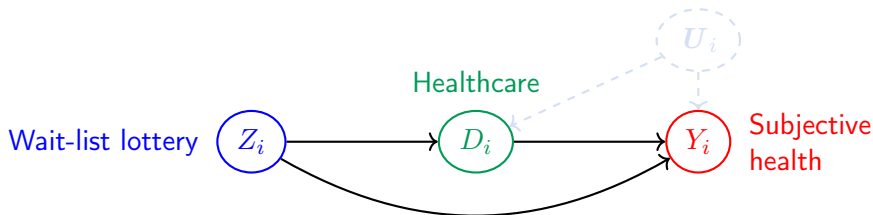
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People chose to visit **healthcare** D_i freely, not randomly assigned...



Conventional CM analyses are misleading in natural experiment settings — I derive non-parametric selection bias terms for CM.

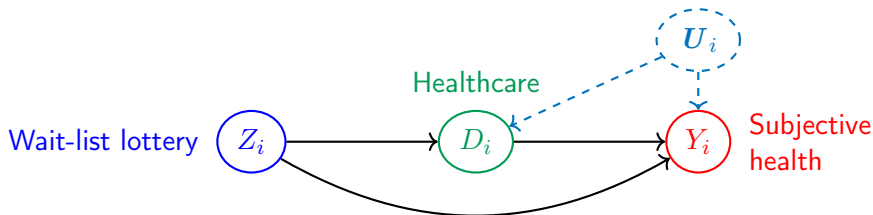
- ADE: CM Estimand = ADE + (Selection Bias + Group difference bias)
- AIE: CM Estimand = AIE + (Selection Bias + Group difference bias)

► ADE biases

► AIE biases

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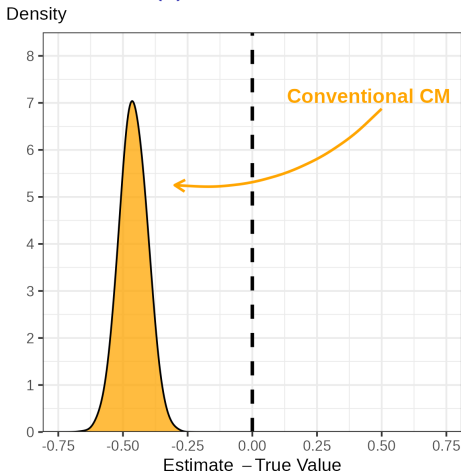
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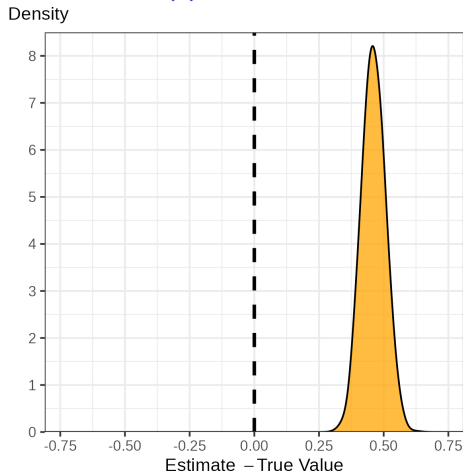
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Simulations with cost-benefit selection-into- D_i , CM estimates are biased.

(a) $\widehat{ADE} - ADE$.



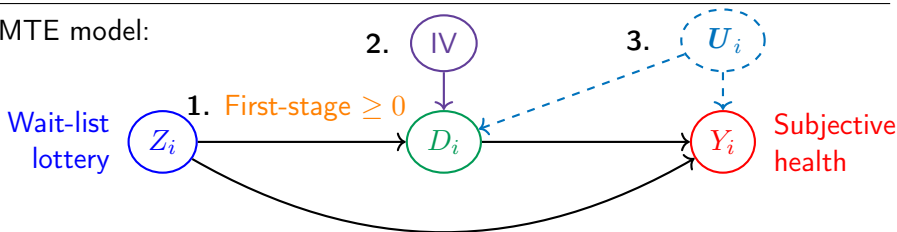
(b) $\widehat{AIE} - AIE$.



3. MTE Model

Conventional CM methods do not identify ADE + AIE in a natural experiment setting, so I build a Marginal Treatment Effect (MTE) model.

MTE model:



MTE assumptions:

- ① Mediator monotonicity
- ② IV for mediator take-up cost
- ③ Selection on mediator benefits.

Intuition: model U_i via mediator MTE to identify ADE + AIE.

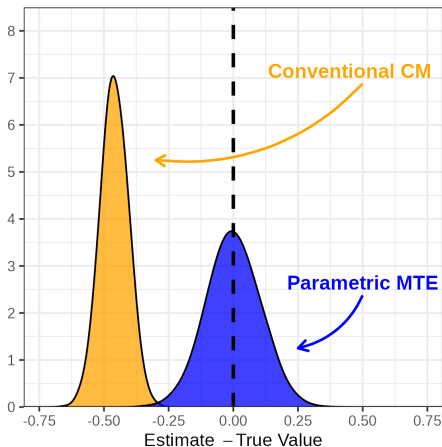
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Simulation with cost-benefit selection-into- D_i , MTE-model corrects for selection bias.

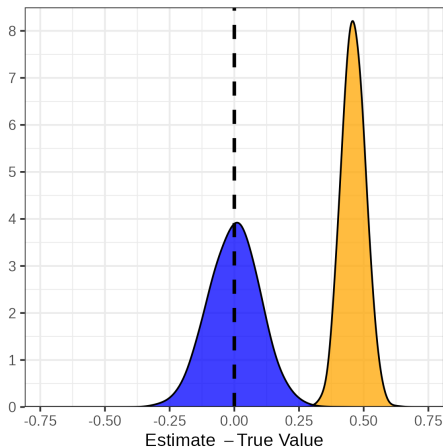
(a) $\widehat{ADE} - ADE$.

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Density



Density



Conclusion

- ① Conventional CM has selection bias in observational (economic) settings
- ② MTE-based model avoids selection bias
- ③ Meaningful indirect (healthcare) and direct (psychological) effects in Oregon Health Insurance Experiment, with wide confidence intervals showing true uncertainty in mechanism explanations.

Questions?